IB MATHEMATICS SL • UNIT 7.01

Topic 7: AA SL Analytical Proofs, Exact Trig & Kinematics

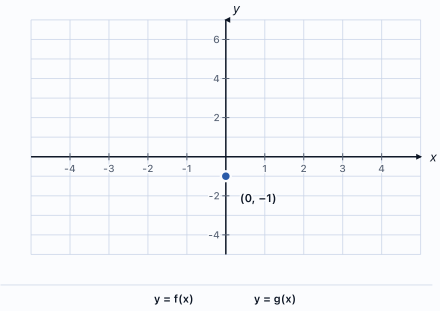
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Topic 7: AA SL Analytical Proofs, Exact Trig & Kinematics Practice Exercises

1 MCQ

Consider the function $f(x) = x^2 - 5x + 6$. What is the value of $f(2)$?



A 0

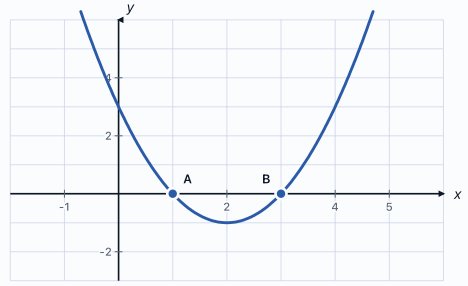
B -2

C 4

D 8

2 MCQ

Consider the quadratic function $f(x) = x^2 - 4x + 3$. The graph of this function is shown below. The points where the graph intersects the x-axis are denoted by A and B. What is the distance between points A and B?



A 1

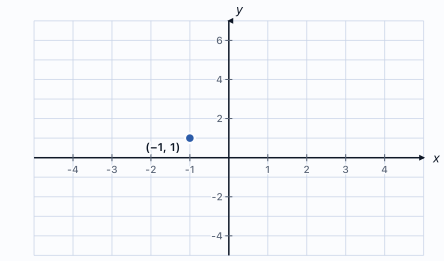
B 3

C 2

D 4

3 MCQ

Consider the function $f(x) = x^2 - 2x + 1$. Determine the y-coordinate of the vertex of the graph of $y = f(x)$.



A 1

B 0

C -1

D 2

4 MCQ

Consider the function $f(x) = 3x^2 - 2x + 5$. Find the value of $f'(1)$.

A 4

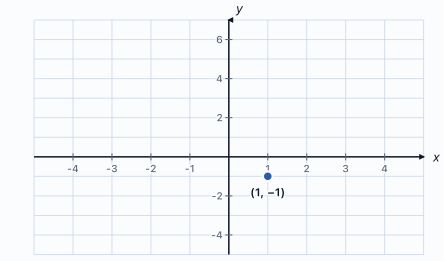
B 5

C 6

D 7

5 MCQ

The graph of the function $f(x) = 2x^2 - 3x + 5$ is shown below. Determine the value of $f'(2)$.



A 7

B 5

C 9

D 2

6 MCQ

Consider the function $f(x) = x^2 - 6x + 5$. Find the x-coordinate of the vertex of the parabola represented by $y = f(x)$.

A 1

B 3

C -1

D -3

7 MCQ

Find $\int x^2 + 4x^3\,dx$.

A $\frac{1}{3}x^3 + x^4 + C$

B $\frac{1}{4}x^4 + \frac{4}{5}x^5 + C$

C $3x^3 + 16x^4 + C$

D $x^2 + 4x^3 + C$

8 MCQ

Find $\int 5x^2 + 4x\,dx$.

A $\frac{5}{3}x^3 + 2x^2 + C$

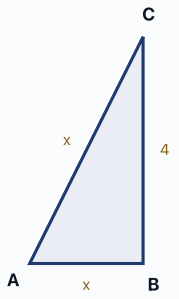
B $\frac{5}{4}x^4 + \frac{4}{3}x^3 + C$

C $15x^3 + 8x^2 + C$

D $5x^2 + 4x + C$

9 MCQ

In the diagram, triangle ABC is shown with coordinates A(1, 1), B(3, 1), and C(3, 5). What is the exact length of the side AC in simplest radical form?



A $2\sqrt{5}$

B 4

C 5

D $4\sqrt{2}$

10 MCQ

Convert 210° to radians, giving your answer as an exact multiple of π.

A $\frac{7\pi}{6}$

B $\frac{5\pi}{3}$

C $\frac{\pi}{2}$

D $\frac{7\pi}{4}$

11 MCQ

Convert 270° to radians, giving your answer as an exact multiple of π.

A $\frac{3\pi}{4}$

B $\frac{\pi}{4}$

C $\frac{3\pi}{2}$

D $\frac{2\pi}{3}$

12 MCQ

Convert 240° to radians, giving your answer as an exact multiple of π.

A $\frac{7\pi}{6}$

B π

C $\frac{4\pi}{3}$

D $\frac{\pi}{3}$

13 MCQ

Convert 330° to radians, giving your answer as an exact multiple of π.

A $\frac{11\pi}{6}$

B $\frac{\pi}{2}$

C $\frac{4\pi}{3}$

D $\frac{\pi}{3}$

14 MCQ

Convert 300° to radians, giving your answer as an exact multiple of π.

A $\frac{5\pi}{3}$

B $\frac{\pi}{3}$

C $\frac{2\pi}{3}$

D $\frac{4\pi}{3}$

15 MCQ

Convert 120° to radians, giving your answer as an exact multiple of π.

A $\frac{5\pi}{4}$

B $\frac{7\pi}{6}$

C $\frac{2\pi}{3}$

D $\frac{3\pi}{2}$

16 MCQ

Convert 150° to radians, giving your answer as an exact multiple of π.

A $\frac{3\pi}{4}$

B $\frac{\pi}{2}$

C $\frac{5\pi}{6}$

D $\frac{7\pi}{4}$

17 MCQ

Convert 90° to radians, giving your answer as an exact multiple of π.

A $\frac{5\pi}{6}$

B $\frac{\pi}{2}$

C $\frac{7\pi}{6}$

D $\frac{2\pi}{3}$

18 MCQ

Convert 45° to radians, giving your answer as an exact multiple of π.

A $\frac{11\pi}{6}$

B $\frac{\pi}{4}$

C π

D $\frac{5\pi}{3}$

19 MCQ

Convert 225° to radians, giving your answer as an exact multiple of π.

A $\frac{11\pi}{6}$

B $\frac{5\pi}{4}$

C $\frac{\pi}{2}$

D $\frac{7\pi}{4}$

20 MCQ

Convert 315° to radians, giving your answer as an exact multiple of π.

A $\frac{\pi}{4}$

B $\frac{3\pi}{4}$

C $\frac{2\pi}{3}$

D $\frac{7\pi}{4}$

21 MCQ

Convert 60° to radians, giving your answer as an exact multiple of π.

A π

B $\frac{5\pi}{3}$

C $\frac{\pi}{3}$

D $\frac{5\pi}{6}$

22 MCQ

Convert 135° to radians, giving your answer as an exact multiple of π.

A $\frac{\pi}{2}$

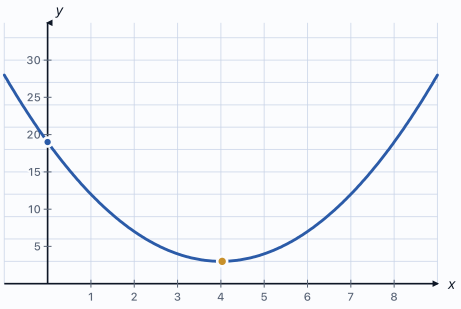
B $\frac{5\pi}{6}$

C $\frac{5\pi}{3}$

D $\frac{3\pi}{4}$

23 MCQ

In AA/AI SL — Functions, Statistics & Analytical Calculus (problem set #998), consider the function $f_{998}(x) = x^2 - 8x + 19$ illustrated on the coordinate axes. Express $f_{998}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{998}(6)$.



A 14

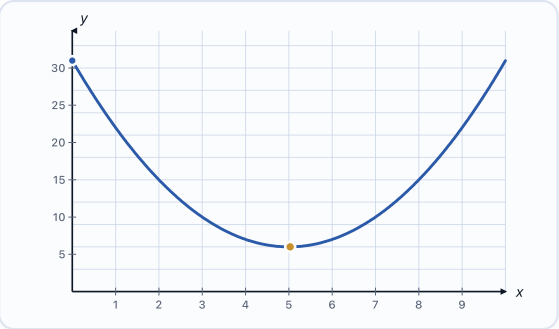
B 7

C 8

D 10

24 MCQ

Considering AA/AI SL — Functions, Statistics & Analytical Calculus (problem set #1001), consider the function $f_{1001}(x) = x^2 - 10x + 31$ illustrated on the coordinate axes. Express $f_{1001}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{1001}(10)$.



A 31

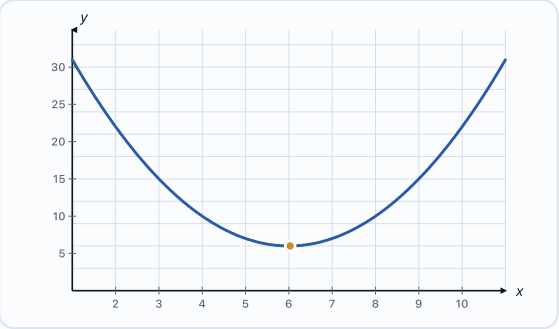
B 32

C 37

D 42

25 MCQ

Examining AA/AI SL — Functions, Statistics & Analytical Calculus (problem set #1004), consider the function $f_{1004}(x) = x^2 - 12x + 45$ illustrated on the coordinate axes. Express $f_{1004}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{1004}(14)$.



A 82

B 73

C 74

D 88